

Name: _____

Date: _____

Trig Definitions

SYNOPSIS

In Note 2, we resolved a vector into a right triangle: the vector itself is the hypotenuse, and its x- and y-components are the two legs, $F = F_x + F_y$. SOH–CAH–TOA is what turns that picture into numbers. Sine, cosine, and tangent are fixed ratios between a right triangle's opposite, adjacent, and hypotenuse sides:

$$\sin\theta = \text{Opp}/\text{Hyp} \Rightarrow \text{Opp} = (\text{Hyp})(\sin\theta)$$

$$\cos\theta = \text{Adj}/\text{Hyp} \Rightarrow \text{Adj} = (\text{Hyp})(\cos\theta)$$

$$\tan\theta = \text{Opp}/\text{Adj}$$

Applied to a resolved vector F at angle θ : F_x ("Adj") = $F \cos\theta$ and F_y ("Opp") = $F \sin\theta$. Once two vectors are each broken into x- and y-components, the components along each axis add like ordinary numbers — that is the whole point of resolving them. The resultant's components recombine into a magnitude and direction: $|C| = \sqrt{C_x^2 + C_y^2}$ and $\theta = \tan^{-1}(\dots)$. The one catch: a reference angle by itself doesn't tell you a component's sign — you must read that off the diagram, based on which quadrant the vector points into.

PERFORMANCE EXPECTATIONS

Compute vector components using $\sin\theta/\cos\theta$; recombine components into magnitude and direction.

Instructions and Questions

1. Label the right triangle below with **Opp**, **Adj**, and **Hyp** relative to the marked angle θ . Then write the three ratios $\sin\theta$, $\cos\theta$, and $\tan\theta$ in terms of those three sides.

2. A vector F is resolved into components F_x and F_y along the x- and y-axes. Which side of the component triangle is F_x — opposite or adjacent to θ ? Which is F_y ? Write the formula for each in terms of F and θ .

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3. Vector **A** has magnitude 10 m/s and a reference angle of 60° from the horizontal, pointing into **Quadrant IV** (right and down). Compute its components.

Q1) $A_x = \underline{\hspace{2cm}}$ m/s $A_y = \underline{\hspace{2cm}}$ m/s

4. Vector **B** has magnitude 5 m/s and a reference angle of 30° from the horizontal, pointing into **Quadrant III** (left and down). Compute its components — remember, the quadrant sets the sign even though 30° itself is a positive angle.

Q2) $B_x = \underline{\hspace{2cm}}$ m/s $B_y = \underline{\hspace{2cm}}$ m/s

5. Vectors A and B are joined head-to-tail (A first, then B), forming resultant **C**. Add the components from Q1 and Q2.

Q3) $C_x = A_x + B_x = \underline{\hspace{2cm}}$ m/s $C_y = A_y + B_y = \underline{\hspace{2cm}}$ m/s

Are your signs for C_x and C_y positive or negative? What does that tell you about which way C points?

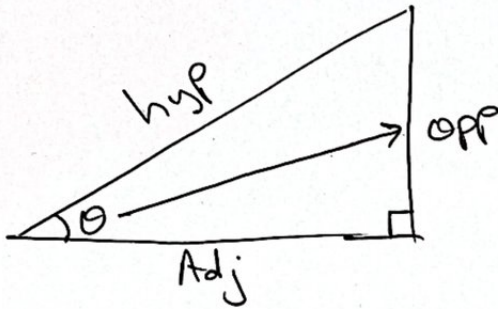
6. Recombine C_x and C_y into the resultant's magnitude and direction.

Q4) $|C| = \sqrt{(C_x^2 + C_y^2)} = \underline{\hspace{2cm}}$ m/s

Q5) $\theta = \tan^{-1}(|C_x| / |C_y|) = \underline{\hspace{2cm}}^\circ$ (measured from the vertical axis, toward +x)

7. Use the reference page below (Dr. Cho's trig-definition sheet) to check your triangle labels and component formulas before moving on.

Trig definition

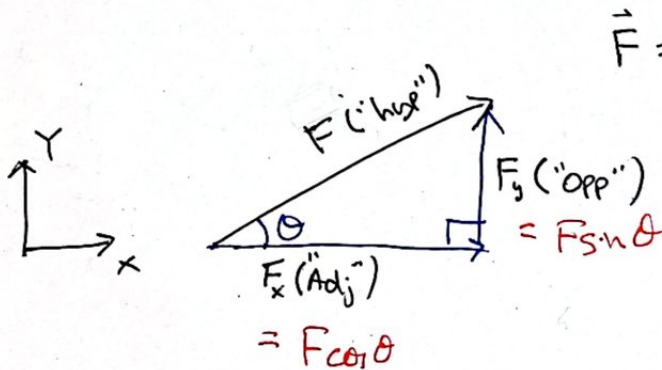


$$\sin \theta = \frac{\text{opp}}{\text{hyp}} \Rightarrow \text{opp} = (\text{hyp})(\sin \theta)$$

$$\cos \theta = \frac{\text{Adj}}{\text{hyp}} \Rightarrow \text{Adj} = (\text{hyp})(\cos \theta)$$

$$\tan \theta = \frac{\text{opp}}{\text{Adj}}$$

Resolving a vector



$$\vec{F} = \vec{F}_x + \vec{F}_y$$

$$\vec{F}_x = F \cos \theta$$

$$\vec{F}_y = F \sin \theta$$

Reference: SOH-CAH-TOA triangle and vector-resolution diagram. Use this to check your triangle labels in Question 1 and 2.